

CITS3002: Lab2-Questions

Exercise 1: Considering **1001** is used as the divisor in the CRC error detection:

- a) Find data delivered to the receiver when the sender transmits **1011001**, assuming that no error occurs.
- b) If the receiver obtains 1100100101, does it detect any error? Explain why.

Exercise 2: Consider two hosts, A and B, are connected by two links (L1, L2). Suppose node A sends a packet with the size of 32 bytes to node B. Answer the following questions (queueing delays can be ignored):

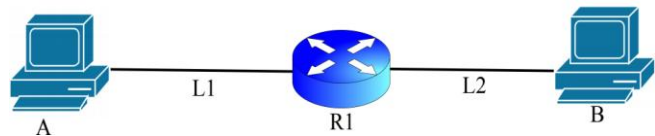
Bandwidth of L1= 1000 bps

Bandwidth of L2= 2000 bps

Length of L1= 10 m

Length of L2= 20 m

Propagation speed of links= 2×10^8 (m/s)



- a) What is the end-to-end delay?
- b) From the moment Router R1 receives the packet, how long does it take for the first bit to reach Host B?

Exercise 3: Hamming code is used for error detection and correction. The received bit pattern is 110110001110. Assume the original data length is 8 bits.

- a) Determine whether a single-bit error has occurred.
- b) If an error exists, identify the position of the erroneous bit.
- c) Provide the corrected bit pattern.